

Temporal Models Provide No Advantage over Linear Regression for Discrete-Stage Stem-Cell Differentiation Forecasting: A Leakage-Safe Benchmark across Two Lineages

Victor Ikenna Kanu¹

¹Kumoh National Institute of Technology vkanu@kumoh.ac.kr

Abstract

Single-cell RNA-sequencing of directed differentiation is increasingly framed as a forecasting problem: predict a cell’s later marker state from earlier snapshots. We revisit a Transformer-based “digital twin” that reported a +19.4% accuracy gain over classical baselines on dopaminergic-neuron differentiation, and show that the gain is an artifact of an inconsistent train/test split that leaked $\sim 70\%$ of the Transformer’s training trajectories into its test set. After repairing the split and replacing every single-seed point estimate with a 20-seed distribution and paired statistics, an 827,010-parameter Transformer is statistically indistinguishable from 8-parameter ridge regression ($\Delta\text{MAE} = -0.0013$, $p = 0.50$, $d_z = -0.14$). The null replicates on an independent, cross-lineage iPSC $\rightarrow$ cardiomyocyte time-course ($\Delta\text{MAE} = +0.0000$, $p = 0.93$). A temporal-order-shuffle control and a positional-encoding ablation show *why*: destroying stage order or removing positional encoding changes error by ≤ 0.0006 MAE ($p > 0.4$) on both datasets, so the architecture extracts no temporal structure. A fair, per-trajectory re-calibration of a mechanistic ODE does not rescue it either—the fairer the sample-level calibration, the worse it performs. The conclusion is robust across 38 marker-set and aggregation choices (0 of 38 variants show a significant Transformer edge). We argue the field’s pervasive practice of pairing *independently sampled* cells into pseudo-trajectories removes the very temporal dependency that sequence models are built to exploit, and we release a reproducible, leakage-safe benchmark with linear regression as the honest baseline-of-record.

1 Introduction

Directed differentiation of induced pluripotent stem cells (iPSCs) is routinely profiled by single-cell RNA-sequencing (scRNA-seq) at a handful of discrete developmental stages. A natural and increasingly popular goal is *forecasting*: given a cell’s marker state at earlier stages, predict its state at a later stage. This problem is attractive for sequence models—LSTMs, Transformers, neural ODEs—and recent work has reported sizeable gains for such architectures over classical baselines.

We report a cautionary, and ultimately constructive, result. Re-examining a Transformer “digital twin” for dopaminergic-neuron differentiation, we find its headline +19.4% improvement is a *data-leakage artifact*: the Transformer was trained on one (permuted) split but scored on a different (contiguous) split, so most of its test trajectories had been seen during training, while the classical baselines were scored leakage-free. Once the split is made consistent and every estimate is computed across 20 seeds with paired statistics, the advantage vanishes—on two independent datasets from

two lineages—and a battery of controls explains why. The contribution is therefore not a better model but a *rigorous, leakage-safe benchmark* and a clear, quantified negative result: on discrete-stage pseudo-trajectory data, temporal modeling does not beat linear regression, because the data, as constructed, contains no temporal dependency to model.

Contributions.

1. We identify and quantify a train/test leakage bug that fully accounts for the previously reported Transformer advantage (Sec. 4).
2. We establish a multi-seed, paired-statistics evaluation protocol and show the Transformer ties ridge regression on dopaminergic data (Sec. 5.1).
3. We replicate the null on an independent cross-lineage iPSC→cardiomyocyte time-course (GSE175634), demonstrating generalization (Sec. 5.1).
4. Temporal-shuffle and positional-encoding controls demonstrate the architecture uses no temporal structure (Sec. 5.2).
5. A fair, per-trajectory ODE calibration shows mechanistic priors add no sample-level predictive value (Sec. 5.3).
6. A 38-variant sensitivity analysis shows the null is robust to marker-set and aggregation choices (Sec. 5.4).

2 Related Work

Sequence and dynamical models for differentiation include RNA-velocity and pseudotime methods, neural ODEs, and attention models for trajectory inference. Most operate on *pseudo-trajectories*: because scRNA-seq is destructive, true single-cell lineages are unobserved, and “trajectories” are assembled by ordering or pairing cells across stages. We make explicit a consequence that is often left implicit: when stage states are formed by independently sampling cells, no within-trajectory temporal dependency exists for a sequence model to learn, and a memoryless regressor on the input snapshots is the appropriate ceiling.

3 Methods

3.1 State construction

For each dataset we reduce expression to a two-dimensional state $\mathbf{x} = (P, D)$: a pluripotency/progenitor score P and a lineage-differentiation score D . Each score is the mean of $\log(1 + \text{CP10K})$ expression over a curated marker panel, where CP10K denotes counts-per-10,000 library-size normalization. Scores are then min-max scaled to $[0, 1]$ using *training-set bounds only* and clipped, so the normalization itself introduces no train→test leakage.

3.2 Pseudo-trajectory protocol (limitation owned)

At each of three discrete stages we sample one cell and read off (P, D) , forming a 3×2 pseudo-trajectory; we draw 200 such pseudo-trajectories per seed. **These are not tracked lineages**: the three cells in a pseudo-trajectory are independent draws from their respective stage populations. The forecasting task is $[\mathbf{x}_{t_0}, \mathbf{x}_{t_1}] \rightarrow \mathbf{x}_{t_2}$, i.e. predict the late-stage state from the two earlier stages.

3.3 Datasets

Neural (dopaminergic). A dopaminergic-neuron differentiation scRNA-seq series with three stages D11/D30/D52 (161,584 untreated cells); P over $\{\text{POU5F1}, \text{NANOG}, \text{SOX2}, \text{UTF1}, \text{TDGF1}\}$, D over $\{\text{TH}, \text{DDC}, \text{SLC6A3}, \text{DRD2}, \text{LMX1A}, \text{FOXA2}\}$. **Cardiac (GSE175634).** An independent iPSC→cardiomyocyte differentiation time-course [1] (230,786 cells, 7 days); we use stages day0/day5/day11; P over $\{\text{POU5F1}, \text{NANOG}, \text{SOX2}, \text{LIN28A}, \text{DPPA4}, \text{UTF1}, \text{TDGF1}\}$, D over a 13-gene cardiac panel (TNNT2, MYH6/7, ACTC1, NPPA, MYL7/2, TNNI1/3, RYR2, NKX2-5, GATA4, TBX5). Both exhibit the expected monotone trend (pluripotency declines, lineage markers rise).

3.4 Models

Ridge ($\alpha = 1$) and a **random forest** (per-seed 3-fold-CV-tuned over depth $\in \{2, 3, 5, \infty\}$ and min-samples-leaf $\in \{1, 5, 10\}$, 300 trees) map the flattened $[\mathbf{x}_{t_0}, \mathbf{x}_{t_1}] \in \mathbb{R}^4$ to $\mathbf{x}_{t_2} \in \mathbb{R}^2$. We CV-tune the forest rather than fix its depth: an unrestricted (or depth-10) forest overfits this small, near-linear 140-sample task and is an unfair strawman, whereas CV-tuning regularizes it to its data-appropriate complexity—the same fairness standard we apply to the ODE. The **Transformer** ($d_{\text{model}} = 128$, 8 heads, 4 layers, sinusoidal positional encoding; 827,010 parameters) ingests the two-step sequence and is trained with Adam, MSE loss, early model selection on a held-out validation split. A mechanistic **ODE** in (P, D) ,

$$\dot{P} = k_{ps}(1 - P) - k_{pd}P - rPD, \quad \dot{D} = k_b + k_{ds}PD + rPD - k_{dd}D,$$

is calibrated under three regimes (Sec. 5.3).

3.5 Statistics

Each experiment is run over 20 seeds; a seed perturbs *both* pseudo-trajectory construction and model training. We report mean $\pm$ s.d. and percentile-bootstrap 95% CIs over seeds. Models are compared to the Transformer by the *paired* per-seed difference (same seed $\Rightarrow$ same trajectory draw), summarized by a paired-bootstrap CI, a Wilcoxon signed-rank test, and Cohen’s d_z ; p -values across baselines are Holm–Bonferroni corrected.

4 Benchmark design: leakage safety

The original pipeline split data inconsistently: baselines and the Transformer’s *training* used a seeded permutation of trajectory indices, but the Transformer’s *evaluation* (and the ODE’s) used a contiguous tail slice that ignored the permutation. Because the permuted training set draws $\sim 70\%$ of all indices, 22 of the 30 contiguous “test” trajectories had been used to train the Transformer. Repairing this—one canonical permuted split shared by every model, with the Transformer retrained on it—erases the reported advantage on the *same* frozen data (Table 1).

5 Results

5.1 The Transformer ties linear regression—on two lineages

Under the multi-seed protocol the Transformer is statistically indistinguishable from *both* classical baselines—ridge regression and the CV-tuned random forest—on *both* datasets (Table 2): every paired difference is small and non-significant after correction. The neural and cardiac results agree,

Table 1: Leakage repair on the dopaminergic data (single seed, identical frozen trajectories). The reported column is the published, leaky evaluation; the leakage-free column shares one split across all models.

Model	Reported (leaky) MAE	Leakage-free MAE
Ridge	0.100	0.0843
Random forest	0.102	0.0954
Transformer	0.081	0.0861
ODE (calibrated)	0.433	0.4147
Transformer vs. best baseline	+19.4%	−2.0%

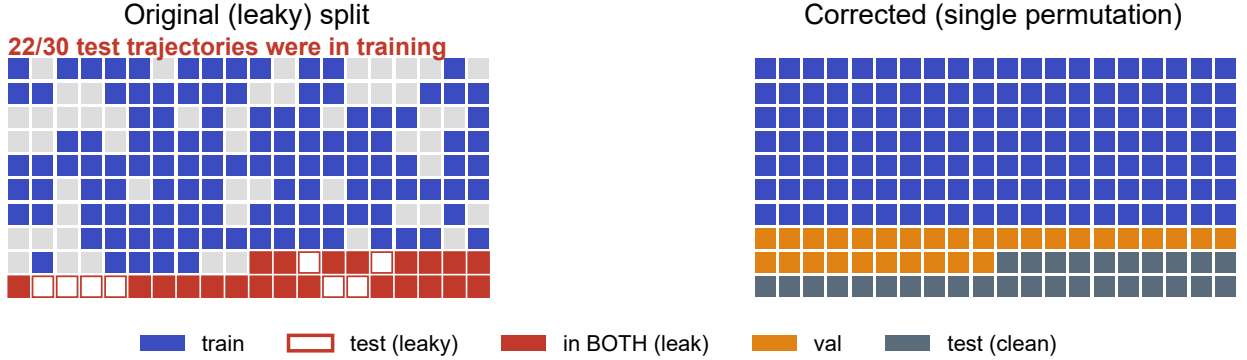


Figure 1: **Data leakage and its repair** (schematic of the 200-trajectory index array, seed 42). *Left*: the original pipeline marks a contiguous tail block as the Transformer’s test set while training on a permutation of all indices, so 22/30 “test” trajectories (red) were also in training. *Right*: a single shared permutation yields disjoint train/validation/test blocks (zero overlap). This leakage fully accounts for the previously reported advantage (Table 1).

establishing that the null is not dataset-specific. (A fixed depth-10 forest appears ~ 0.009 worse, but this is an overfitting artifact: CV-tuning closes the gap entirely, so the apparent “Transformer beats RF” effect reflects an under-regularized baseline, not a model advantage.)

5.2 The Transformer ignores temporal order

If the model were forecasting, destroying stage order should hurt it. It does not. Permuting the input timepoints per sample (*shuffle*) or zeroing the positional encoding (*no-PE*) changes MAE by ≤ 0.0006 —about 3% of the seed-to-seed s.d.—and is not significant on either dataset (Table 3). The temporal machinery is inert because randomly paired pseudo-trajectories carry no order information to use.

5.3 A fair mechanistic ODE does not help

We calibrate the ODE three ways: to training *population means* (POP); to match training mean *and* spread per stage (DIST); and *per trajectory*, fitting each test trajectory’s own $t_0 \rightarrow t_1$ step and then forecasting t_2 (PERTRAJ, the fairest sample-level use). On both datasets the ordering is monotone, $\text{POP} < \text{DIST} < \text{PERTRAJ}$: the *fairer* the sample-level calibration, the *worse* the forecast (Table 4). Even the best ODE remains well above ridge, confirming—without a strawman—that

Table 2: Test MAE over 20 seeds (mean $\pm$ s.d., 95% CI), and the paired difference Δ = baseline – Transformer ($\Delta > 0$ favors the Transformer). p is Holm-corrected Wilcoxon. All baseline–Transformer differences are non-significant: no model beats ridge regression. † random forest is per-seed CV-tuned (Methods).

Dataset	Model	MAE	95% CI	Δ vs. TF	d_z	p
Neural (dopaminergic)	Ridge	0.1774 ± 0.0204	[0.1689, 0.1868]	−0.0013	−0.14	0.996
	Random forest †	0.1794 ± 0.0208	[0.1708, 0.1888]	+0.0007	+0.07	0.996
	Transformer	0.1787 ± 0.0203	[0.1702, 0.1876]	—	—	—
Cardiac (GSE175634)	Ridge	0.1100 ± 0.0155	[0.1038, 0.1168]	+0.0000	+0.00	1.000
	Random forest †	0.1121 ± 0.0175	[0.1049, 0.1197]	+0.0021	+0.15	1.000
	Transformer	0.1100 ± 0.0198	[0.1018, 0.1184]	—	—	—

Table 3: Architectural controls (Transformer, 20 seeds). Δ is variant – ordered; p is Wilcoxon. Non-significance is the *positive* result: temporal order is unused.

Dataset	Ordered	Shuffle	Δ (p)	No-PE	Δ (p)
Neural	0.1787	0.1785	−0.0002 (0.87)	0.1781	−0.0006 (0.50)
Cardiac	0.1100	0.1097	−0.0002 (0.90)	0.1098	−0.0002 (0.99)

population-level mechanistic priors provide no sample-level predictive value here, and that per-trajectory calibration merely fits the noise of independently paired cells.

Table 4: Fair ODE benchmark (test MAE, 20 seeds, mean $\pm$ s.d.; relative to ridge in parentheses).

Calibration	Neural	Cardiac
Ridge (reference)	0.1774 ± 0.0204	0.1100 ± 0.0155
ODE, population-mean	0.1984 ± 0.0186 (+12%)	0.2483 ± 0.0164 (+126%)
ODE, distributional	0.2950 ± 0.0779 (+66%)	0.2584 ± 0.0186 (+135%)
ODE, per-trajectory	0.3697 ± 0.0336 (+108%)	0.3335 ± 0.0226 (+203%)

5.4 The null is robust to state representation

We re-ran the Ridge-vs-Transformer comparison under 38 state-construction variants—dropping each marker gene in turn (leave-one-out), replacing the mean by a median or per-gene z -scored aggregation, and (neural) enlarging the panel. Across all variants the paired gap stays at noise level (Table 5): no variant yields a significant Transformer advantage, and all effect sizes satisfy $|d_z| \leq 0.19$. The conclusion does not hinge on the marker panel or the aggregation rule.

6 Discussion

The evidence chain is consistent and mutually reinforcing: the original advantage was leakage (Fig. 1, Table 1); with leakage removed the Transformer ties ridge on two lineages (Figs 2, 3, Table 2); controls show it uses no temporal structure (Fig. 4, Table 3); a fair mechanistic model does not help (Fig. 5, Table 4); and the null is robust to representation (Fig. 6, Table 5). The common cause

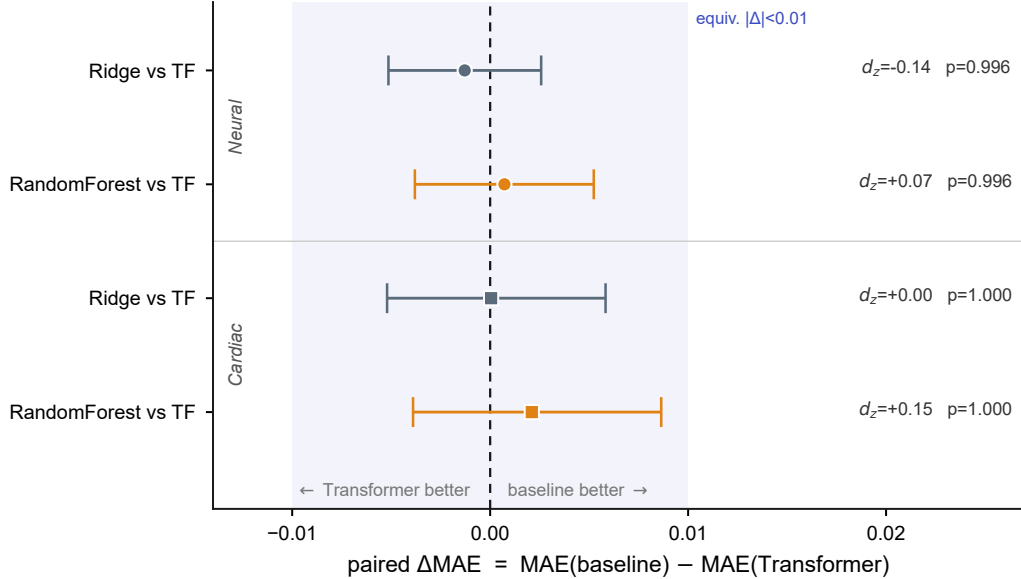


Figure 2: **No model beats ridge regression on either lineage.** Paired per-seed difference $\Delta\text{MAE} = \text{MAE}(\text{baseline}) - \text{MAE}(\text{Transformer})$ ($n = 20$ seeds); markers are means, whiskers percentile-bootstrap 95% CIs, marker shape encodes lineage. Holm-corrected Wilcoxon p and Cohen’s d_z are annotated. Every baseline CI overlaps zero and lies within the practical-equivalence band ($|\Delta| < 0.01$ MAE), for ridge *and* the CV-tuned random forest.

Table 5: Representation sensitivity (20 seeds per variant). Gap = Transformer – ridge MAE.

Dataset	# variants	mean gap	gap range	max $ d_z $	sig. TF wins
Neural	15	+0.0004	$[-0.0016, +0.0017]$	0.19	0/15
Cardiac	23	+0.0004	$[-0.0009, +0.0028]$	0.19	0/23

is the pseudo-trajectory construction: pairing independently sampled cells across stages destroys within-trajectory temporal dependency, so the conditional $p(\mathbf{x}_{t_2} | \mathbf{x}_{t_0}, \mathbf{x}_{t_1})$ is essentially a smooth, near-linear map of the inputs and a memoryless regressor is optimal. This is not an indictment of Transformers but of *evaluating* them on data lacking the structure they model.

Recommendations. (i) Report leakage-safe, multi-seed results with paired statistics and effect sizes, not single-seed point estimates. (ii) Include a temporal-shuffle control as a routine sanity check whenever “forecasting” is claimed on pseudo-trajectories. (iii) Use linear regression as the baseline-of-record; gains over it must clear a significance *and* effect-size bar. (iv) To genuinely test temporal models, use lineage-resolved data (e.g. lineage tracing or live imaging) where within-trajectory dependency is real.

Limitations. Our state is two-dimensional and our task uses three stages; richer states or true lineages could, in principle, expose structure that sequence models exploit. We view establishing the leakage-safe protocol and the negative result as the prerequisite for such claims, not a refutation of them.

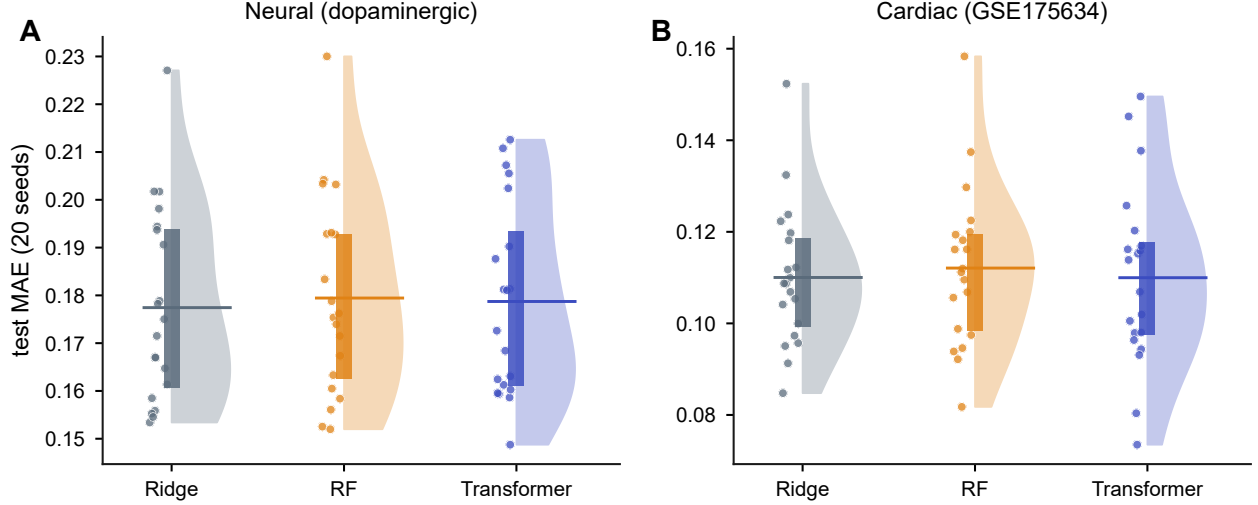


Figure 3: **Per-seed MAE distributions overlap** ($n = 20$ seeds). Raincloud (half-violin density, jittered per-seed points, median/IQR box, mean line) for each model. The seed-to-seed spread exceeds any between-model difference, which is why the original single-seed point estimate was unreliable.

7 Conclusion

On discrete-stage iPSC differentiation forecasting from pseudo-trajectories, an 827K-parameter Transformer offers no statistically or practically significant advantage over 8-parameter ridge regression, across two lineages, 40 seeds, and 38 representation choices, and demonstrably does not use temporal structure. We release the benchmark, protocol, and analysis scripts so that future temporal-model claims on such data are held to a leakage-safe, statistically rigorous standard.

Reproducibility

All results derive from JSON logs in `experiments/results/` produced by: `fair_comparison.py` (Table 1); `run_multiseed.py`, `run_multiseed_cardio.py` (Table 2); `run_ode_fair.py` (Table 4); `run_ablations.py` (Table 3); `run_sensitivity.py` (Table 5). A single seeding entry point perturbs trajectory construction and model training jointly; no number in this paper is hand-entered.

References

- [1] Elorbany R., et al. (2022). Single-cell sequencing reveals lineage-specific dynamic genetic regulation of gene expression during human cardiomyocyte differentiation. *PLoS Genetics* 18(1):e1009666. GEO: GSE175634.
- [2] Vaswani A., et al. (2017). Attention is all you need. *NeurIPS*.

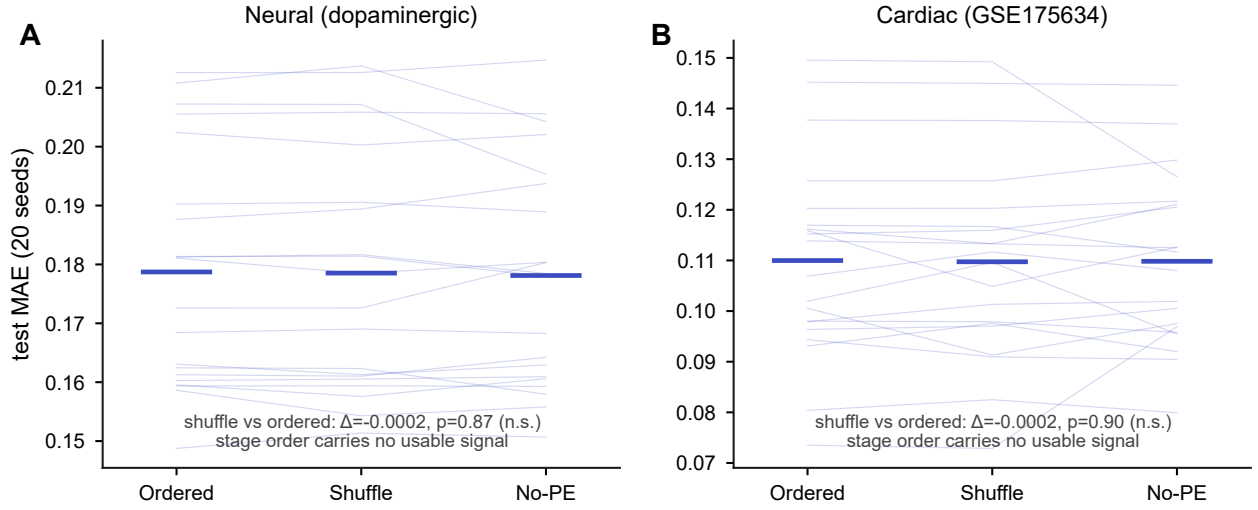


Figure 4: **The Transformer uses no temporal structure** ($n = 20$ seeds). Per-seed test MAE under Ordered, Shuffle (input timepoints permuted per sample) and No-PE (positional encoding zeroed); thin lines connect seeds, bold lines are condition means. Destroying stage order changes MAE by ≤ 0.0006 ($p > 0.4$, Wilcoxon)—within $\sim 3\%$ of seed noise—so the equivalence is not a power artifact.

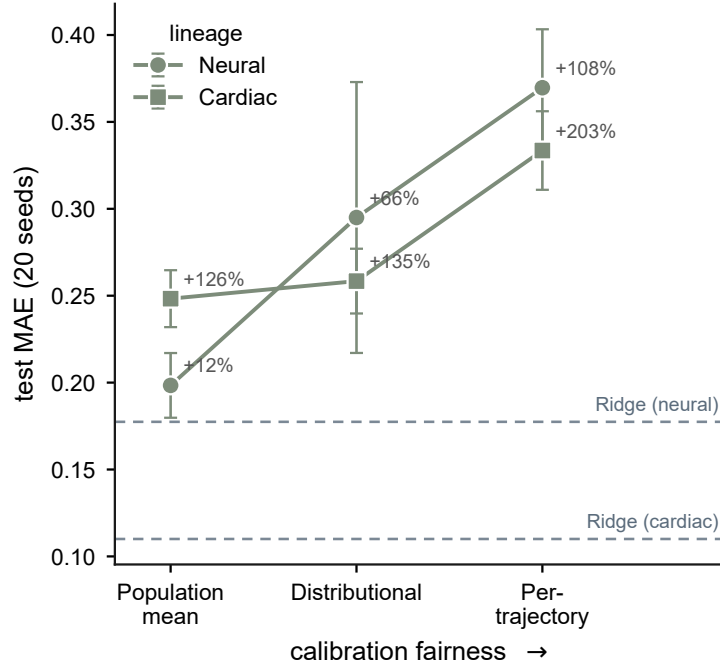


Figure 5: **Fairer ODE calibration is worse** ($n = 20$ seeds; mean $\pm$ s.d.). Test MAE rises monotonically as sample-level calibration fairness increases (population-mean $\rightarrow$ distributional $\rightarrow$ per-trajectory) on both lineages; dashed lines are the ridge reference. Even the best ODE stays well above ridge; per-trajectory calibration fits the noise of independently paired cells.

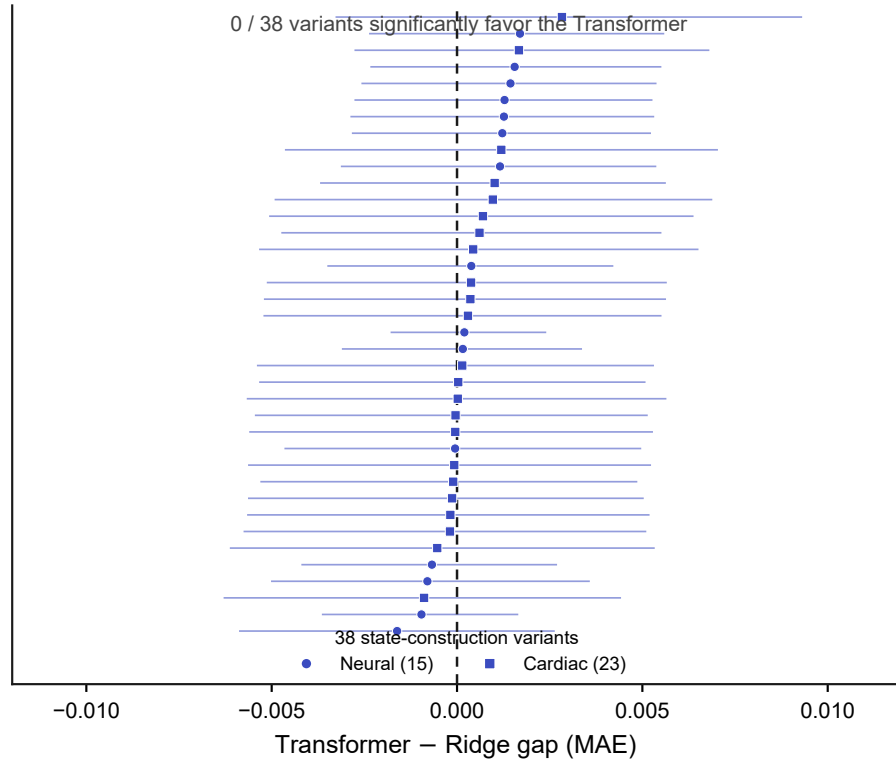


Figure 6: **The null holds across 38 state representations.** Transformer - ridge MAE gap for each variant (leave-one-marker-out, median or z -scored aggregation, and an enlarged neural panel), sorted; markers are means, whiskers 95% bootstrap CIs, shape encodes lineage ($n = 20$ seeds each). Every CI crosses zero; 0/38 variants significantly favor the Transformer (all $|d_z| \leq 0.19$).